Then, $OR = x$, $OS = y$, and $QP = z$.

Thus, $\overrightarrow{OR} = x\hat{i}$, $\overrightarrow{RQ} = \overrightarrow{OS} = y\hat{j}$, and $\overrightarrow{QP} = z\hat{k}$

$$\overrightarrow{OP} = \vec{r} = \overrightarrow{OQ} + \overrightarrow{QP} = \overrightarrow{OR} + \overrightarrow{RQ} + \overrightarrow{QP} = x\hat{i} + y\,\hat{j} + z\hat{k}\,.$$

That is $\overrightarrow{OP} = \vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

This $\overrightarrow{OP}$ vector is called the position vector of $P$ with respect to the origin $O$ in three dimension.

In the triangle $ORQ$,

$$OQ^2 = OR^2 + RQ^2 \text{ (how?)}$$

and in the triangle $OQP$,

$$OP^2 = OQ^2 + QP^2.$$

Thus $OP^2 = OQ^2 + QP^2 = OR^2 + RQ^2 + QP^2 = x^2 + y^2 + z^2$

and hence $|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$, that is $|\vec{r}| = r = \sqrt{x^2 + y^2 + z^2}$

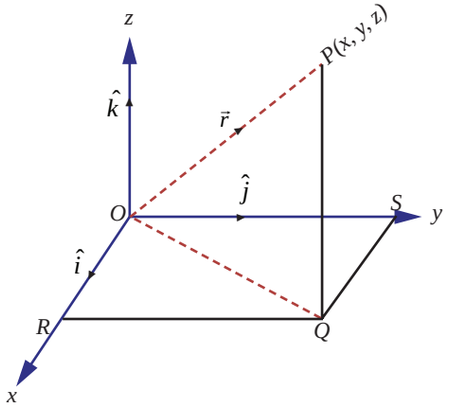



*Fig. 8.31*

### Components of vector joining two points

Let us find the components of the vector joining the point $(x_1, y_1)$ to $(x_2, y_2)$.

Let $A$ and $B$ be the points $(x_1, y_1)$ and $(x_2, y_2)$. Let $P$ be the point $(x_2 - x_1, y_2 - y_1)$. Then $\overrightarrow{AB} = \overrightarrow{OP}$. The components of $\overrightarrow{OP}$ are $(x_2 - x_1)\hat{i}$ and $(y_2 - y_1)\hat{j}$. Hence the components of $\overrightarrow{AB}$ in the directions of $x$ and $y$ axes are $(x_2 - x_1)\hat{i}$ and $(y_2 - y_1)\hat{j}$.

Similarly if $A$ and $B$ are the points $(x_1, y_1, z_1)$ and $(x_2, y_2, z_2)$, then the components of $\overrightarrow{AB}$ in the directions of $x$, $y$ and $z$ axes are $(x_2 - x_1)\hat{i}$, $(y_2 - y_1)\hat{j}$, and $(z_2 - z_1)\hat{k}$.

## 8.6.3 Matrix representation of a vector

A vector with three components can be visualised as either a row or column matrix as $[x, y, z]$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ respectively.

Thus any vector $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ can be obtained from $[a_1 \;\; a_2 \;\; a_3]\begin{bmatrix} \hat{i} \\ \hat{j} \\ \hat{k} \end{bmatrix} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} = \vec{A}$ .

Hence addition of vectors and multiplication of a vector by a scalar can be defined as follows.

If $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ then $\vec{A} + \vec{B} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} a_1 + b_1 \\ a_2 + b_2 \\ a_3 + b_3 \end{bmatrix}$ resulting in

$\vec{A} + \vec{B} = (a_1 + b_1)\hat{i} + (a_2 + b_2)\hat{j} + (a_3 + b_3)\hat{k}$ .

Also $k\vec{A} = k\begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} ka_1 \\ ka_2 \\ ka_3 \end{bmatrix}$ yielding



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